Lecture 8: Hypothesis Testing: Z-score and Chi-Square

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1. Agenda and Announcements

- Today

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  - Hypothesis Testing: Z-Score and Chi-Square
  - Top Hat Quiz
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- Next class

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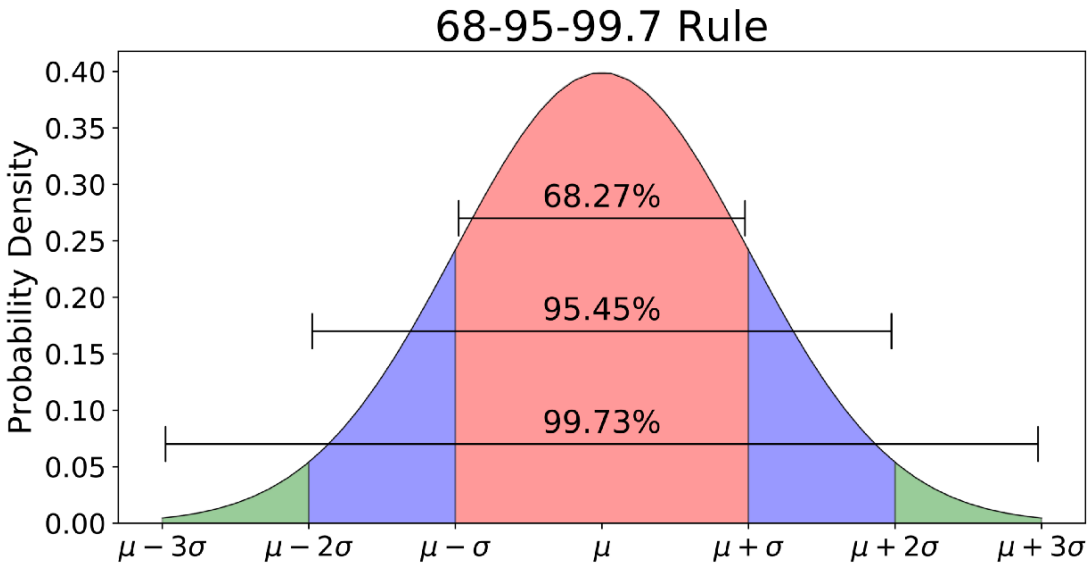
  - Lab 4 completion
  - Lab 5
```

2. Hypothesis testing Z-Scores

2.1 What is a Z-Score?

- A z-score tells us how many standard deviations a value is from the mean.
- Positive z-scores are above the mean
- Negative z-scores are below the mean
- A z-score is the **precise** representation of the 68-95-99 Rule

2.2 The 68-95-99.7 Rule

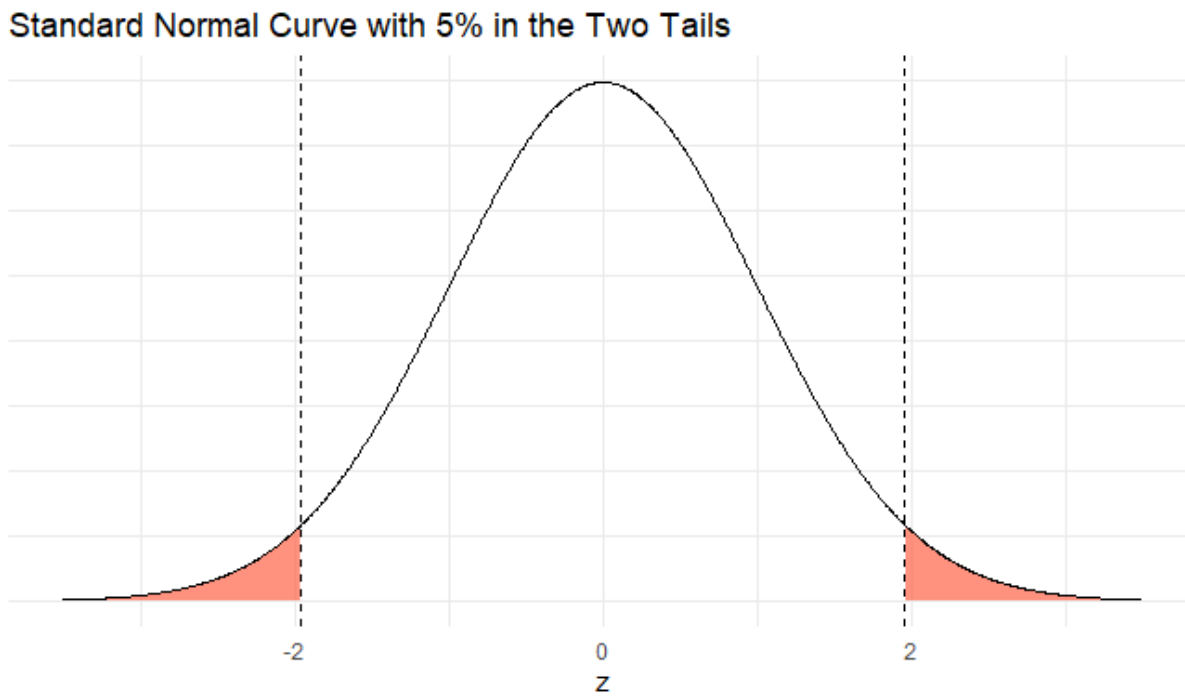


68-95-99.7 rule

2.3 What is a Z-Score? (cont)

- The actual 95% area under the normal curve is between -1.96 and +1.96 standard deviations from the mean
- The actual 95% area is from z = -1.96 to z = 1.956
- If the area under the curve is 95%, then 5% is outside it in the **two tails**

2.4 The tails



2.5 Two-tail test

- In a hypothesis test, we are testing the probability that the event is not due to random chance
- We set a threshold (usually α = 0.05) for how unlikely a result must be before we reject the null hypothesis
- In a two-tail test, we split α into both tails: α/2 = 0.025 in each tail
- If our test statistic falls in either tail (beyond ±1.96), we reject the null hypothesis
- This corresponds to a probability less than 5% or p < .05

2.6 Z-Test: Formula

Formula:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ_0 = hypothesized population mean (from null hypothesis)
- σ = population standard deviation
- n = sample size

2.7 Z-Test: Steps

Steps:

1. State your null hypothesis ($H_0: \mu = \mu_0$) and alternative hypothesis (H_a)
2. Calculate the test statistic using the formula above
3. For a two-tail test at $\alpha = 0.05$, the critical values are $z = \pm 1.96$

2.8 Z-test: Decision rule

Two tail:

- **Reject** H_0 if $z > 1.96$ or $z < -1.96$
- **Fail to reject** H_0 if $-1.96 \leq z \leq 1.96$
- Shorthand: reject if $|z| > 1.96$

2.9 Z-tail: Right tail decision rule

Right-tail test ($H_a: \mu > \mu_0$)

- Critical value: $z_{critical} = 1.645$
- Decision rule:
  - **Reject** H_0 if $z > 1.645$
  - **Fail to reject** H_0 if $z \leq 1.645$
  - Shorthand: reject if z is in the **right tail** beyond 1.645

2.10 Z-tail: Left-tail decision rule

Left-tail test ($H_a: \mu < \mu_0$)

- Critical value: $z_{critical} = -1.645$
- Decision rule:
  - **Reject** H_0 if $z < -1.645$
  - **Fail to reject** H_0 if $z \geq -1.645$
  - Shorthand: reject if z is in the **left tail** beyond -1.645

2.11 What is a Crosstab?

- A **crosstab** (contingency table) shows counts for two categorical variables in rows and columns
- Each cell contains the **frequency** of cases with that combination of categories

2.12 Example Crosstab

Example structure (2×3 table):

	Category 1	Category 2	Category 3
Group A	?	?	?
Group B	?	?	?

2.13 Example: Observed Counts Only

Research Question: Is there an association between gender and support for a new campus policy?

Observed data from 200 students:

	Support	Oppose
Female	60	40
Male	45	55

2.14 Example: Add Row and Column Totals

Fill in the marginals (row totals, column totals, and grand total):

	Support	Oppose	Row Total
Female	60	40	___
Male	45	55	___
Column Total	___	___	**___**

2.15 Example: Completed Observed Table

	Support	Oppose	Row Total
Female	60	40	100
Male	45	55	100
Column Total	105	95	**200**

2.16 Example: Calculate Expected Counts

Formula: $E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{\text{grand total}}$

Calculate expected count for each cell:

- Female & Support: $E_{11} = \frac{100 \times 105}{200} = ___$
- Female & Oppose: $E_{12} = \frac{100 \times 95}{200} = ___$
- Male & Support: $E_{21} = \frac{100 \times 105}{200} = ___$
- Male & Oppose: $E_{22} = \frac{100 \times 95}{200} = ___$

2.17 Example: Expected Counts Table

	Support	Oppose	Row Total
Female	52.5	47.5	100
Male	52.5	47.5	100
Column Total	105	95	**200**

These are the counts we would expect if there were NO association

2.18 Example: Calculate (O - E) for Each Cell

For each cell, compute: Observed - Expected

	Support	Oppose
Female	60 - 52.5 = ___	40 - 47.5 = ___
Male	45 - 52.5 = ___	55 - 47.5 = ___

2.19 Example: Differences (O - E)

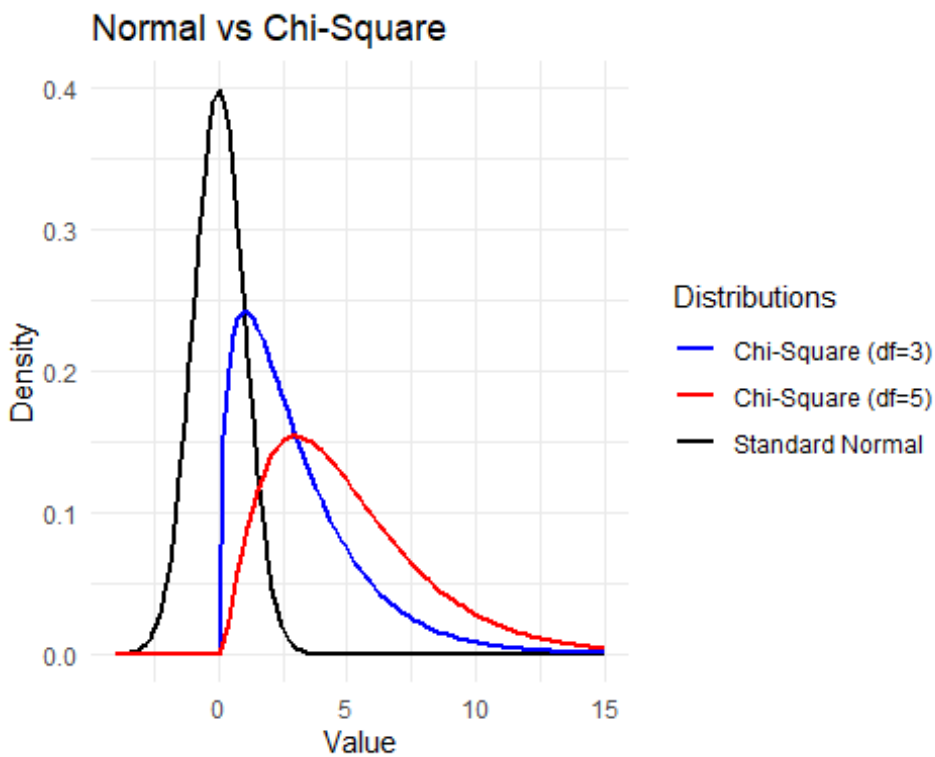
	Support	Oppose
Female	+7.5	-7.5
Male	-7.5	+7.5

Note the symmetry: Differences sum to zero in each row and column

2.20 Where did we run into this before?

- Variance
- When we talked about computing confidence intervals, squaring deviations meant we couldn't use the normal distribution
- That meant we couldn't use a z-score there
- What distribution did we have to use?

2.21 Distribution Comparison



2.22 χ^2 instead of z-score

- Here we will use the χ^2 statistic instead of the z-score
- The z-score is always based on the **same** normal distribution
- The chi-square is not just one curve: we pick a specific χ^2 **distribution** based on the degrees of freedom

2.23 What is degrees of freedom?

- Chi-square shows up whenever we add up **squared differences between what we observe and what we expect**
- These are also called **squared residuals**
- The **degrees of freedom** tell us how many **independent pieces** of information went into that total
- In a crosstab, fixing the row and column totals means not every cell can vary freely, so the df is $(r-1)(c-1)$
- So instead of one universal z table, we have a family of χ^2 tables—one curve for each degrees-of-freedom value.

2.24 χ^2 Critical Values ($\alpha = 0.05$)

df	χ^2 critical ($\alpha = 0.05$)
1	3.84
2	5.99
3	7.82
4	9.49
5	11.07
6	12.59

2.25 Example: next step

Calculate $\frac{(O-E)^2}{E}$ for Each Cell

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Female & Support: $\frac{(7.5)^2}{52.5} = \frac{56.25}{52.5} = __$

Female & Oppose: $\frac{(-7.5)^2}{47.5} = \frac{56.25}{47.5} = __$

Male & Support: $\frac{(-7.5)^2}{52.5} = \frac{56.25}{52.5} = __$

Male & Oppose: $\frac{(7.5)^2}{47.5} = \frac{56.25}{47.5} = __$

2.26 Example: Chi-Square Components

Female & Support: $\frac{56.25}{52.5} = 1.071$

Female & Oppose: $\frac{56.25}{47.5} = 1.184$

Male & Support: $\frac{56.25}{52.5} = 1.071$

Male & Oppose: $\frac{56.25}{47.5} = 1.184$

Sum all four: $\chi^2 = 1.071 + 1.184 + 1.071 + 1.184 = __$

2.27 Example: Chi-Square Test Statistic

$$\chi^2_{calc} = 4.51$$

Now we need to compare this to the critical value

2.28 Example: Find Degrees of Freedom

$$df = (r - 1)(c - 1)$$

- Number of rows: $r =$ __
- Number of columns: $c =$ __

$$df = (__ - 1)(__ - 1) = __$$

2.29 Example: Degrees of Freedom

$$df = (r - 1)(c - 1)$$

- Number of rows: $r = 2$
- Number of columns: $c = 2$

$$df = (2 - 1)(2 - 1) = 1$$

2.30 Example: Critical Value

From chi-square distribution table:

- $df = 1$
- $\alpha = 0.05$
- $\chi^2_{critical} = 3.841$

2.31 Example: Make Decision

Decision rule: Reject H_0 if $\chi^2_{calc} > \chi^2_{critical}$

Our values: - $\chi^2_{calc} = 4.51$ - $\chi^2_{critical} = 3.841$

Decision: _____

2.32 Example: Conclusion

Since $4.51 > 3.841$**, we REJECT** H_0

Conclusion: - There **is** a statistically significant association between gender and support for the policy - The pattern in our sample is unlikely to be due to chance alone - Females show higher support (60%) than males (45%)

2.33 Berkeley Graduate Admissions 1973

Did America's most progressive graduate school discriminate on gender in admissions?

```
, , Dept = A
```

```
          Gender
Admit      Male Female
  Admitted  512     89
  Rejected  313     19

, , Dept = B

          Gender
Admit      Male Female
  Admitted  353     17
  Rejected  207      8

, , Dept = C

          Gender
Admit      Male Female
  Admitted  120    202
  Rejected  205    391

, , Dept = D

          Gender
Admit      Male Female
  Admitted  138    131
  Rejected  279    244

, , Dept = E

          Gender
Admit      Male Female
  Admitted   53     94
  Rejected  138    299

, , Dept = F

          Gender
Admit      Male Female
  Admitted   22     24
  Rejected  351    317

                Dept   A   B   C   D   E   F
Admit    Gender
Admitted Male        512 353 120 138  53  22
         Female       89  17 202 131  94  24
Rejected Male        313 207 205 279 138 351
         Female       19   8 391 244 299 317
```

2.34 1973 Berkeley Example Data (Overall)

Overall admissions (all departments combined)

	Admitted	Rejected	Total
Male	44	56	100
Female	30	70	100
Total	74	126	200

2.35 1973 Berkeley Example Data (By Department)

Admissions by department (simplified)

Dept	Gender	Admitted	Rejected	Total
A	Male	40	60	100
	Female	45	55	100
B	Male	30	70	100
	Female	36	64	100
C	Male	30	70	100
	Female	15	85	100

2.36 For this week's quiz points (25 points)

- Complete the crosstabs hypothesis test for the table with admissions by department. (That is a single test with 6 rows by 2 columns)
- Do a crosstab for each department. (That is 3 tests with 2 rows by 2 rows)
- Do this by hand and give your conclusions.
- More on the next slide

2.37 Part 2

- Bring this to class Thursday or turn it in my mailbox in the Political Science Department office (PGH 447)
- Mostly completion - if you make errors, we will discuss
- Read these and answer three questions in Canvas "March 23-26 Quiz"

```
  - https://setosa.io/simpsons/
  - https://statisticsbyjim.com/basics/simpsons-paradox/
```

- We will revisit using R in lab on Thursday!

- I will post Thursday lab Monday or Tuesday evening with recording

2.38 Authorship, License, Credits

- 68-95-99 rule graphic Source:https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2

- Author: Tom Hanna

- Website: tomhanna.me

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